

National Technical University of Athens
School of Electrical and Computer Engineering
Division of

Thesis Title

Thesis subtitle

Diploma Thesis

of

IOANNIS PAPADOPOULOS



Supervisor: Ioannis Papadopoulos
Assistant Professor

Athens, February 2021



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Approved by the examination committee on 26th February 2021.

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Athens, February 2021



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Abstract

A peer-to-peer system is a set of autonomous computing nodes (the peers) which cooperate in order to exchange data. The peers in the peer-to-peer systems that are widely used today, rely on simple keyword selection in order to search for data. The need for richer facilities in exchanging data, as well as, the evolution of the Semantic Web, led to the evolution of the schema-based peer-to-peer systems. In those systems every node uses a schema to organize the local data. So there are two ways in order for data search to be feasible. The first but not so flexible way implies that every node uses the same schema. The second way gives every node the flexibility to choose a schema according with its needs, but on the same time requires the existence of mapping rules in order for queries to be replied. This way though, doesn't offer automatic creation and dynamic renewal of the mapping rules which would be essential for peer-to-peer systems.

This diploma thesis aims to the development of a schema-based peer-to-peer system that allows a certain flexibility for schema selection and on the same time enables query transformation without the use of mapping rules. The peers use RDF schemas that are subsets (views) of a big common schema called global schema.

Keywords

Peer-to-peer, Schema-based peer-to-peer, Semantic Web, RDF/S, RQL, Jxta

to my parents

Acknowledgements

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Chapter 1

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Introduction

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Chapter Name

Cite test .[1].

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1.<?xmlversion="1.0"?>
2.<rdf:RDFxmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
3.      xmlns:dc="http://purl.org/dc/elements/1.1/"
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7.     <dc:language>en</dc:language>
8.     <dc:creator rdf:resource="http://www.example.org/staffid/85740"/>
9.   </rdf:Description>
10.</rdf:RDF>
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Part I

Epilogue

Bibliography

[1] M. Goossens, F. Mittelbach και A. Samarin. *The LaTeX Companion*. Addison-Wesley, Reading, Massachusetts, 9η έκδοση, 1993.

List of Abbreviations

BPF	Band Pass Filter